One story, seven minutes

You've done three sections. They're not seven separate topics; they're one story with seven scenes. Learn the story, and the facts hang off it.

THE STORY (memorize this paragraph; everything else is detail):
Space gives you nothing for free. **Getting there** is falling and missing: go sideways fast enough and the ground curves away under you. **Where you end up** depends on how high you go: low is fast and sharp but sees little, high is slow and sees everything, and one exact height hangs still. **Describing** any orbit takes six numbers, which are really six ways to describe a hula hoop. **Staying alive** up there means doing everything the air used to do for you, because the vacuum does none of it, and the thin air that's left eats your plastic. **Going somewhere else** is a fuel bill: fast is expensive, the lazy half-ellipse is cheap, and the only free lunch is bouncing off a moving planet.



1	2	3	4	5	6	7
orbit = fall + miss S1 · S3	LEO · MEO · GEO S1 · S3	a e i Ω ω ν S3	vacuum, outgassing S2	layers, atomic O S2	Lambert, Hohmann S1 · S3	flyby, Rosetta S1 · S3

The 7 minutes at a glance

0:00–0:45	0:45–1:45	1:45–2:45	2:45–3:45	3:45–4:30	4:30–5:45	5:45–6:30	6:30–7:00
HOOK falling & missing	SHELVES LEO MEO GEO	HULA HOOP 6 numbers	FREE SERVICES vacuum	AIR BITES layers, atomic O	FUEL BILL Hohmann	FREE LUNCH flyby	LANDING "nothing's free"

Page 2–3: each scene re-taught like you're in 5th grade. Page 4: the minute-by-minute script. Page 5–6: our quiz and answers. Page 7: how to rehearse, and your cue card.

Scenes 1–4: getting there and staying alive

🏀 Scene 1 · Falling and missing

Throw a ball sideways. It falls and lands. Throw it harder, it lands farther. Now throw it so hard that by the time it falls, the ground has curved away underneath it. It never lands. It just keeps falling *around* the Earth. That's an orbit. The space station isn't floating; it's falling, and missing, 16 times a day.

> **SAY: "An orbit is falling and missing. Go sideways fast enough and you never hit the ground."**

Draw: a mountain, three throws, the third one wrapping around.
Friend asks "so why do astronauts float?" → Because they're falling too, together with the station. Same as the drop on a roller coaster, but forever.

🏢 Scene 2 · Three shelves

Satellites live on three shelves. **Low shelf (LEO):** 200–2,000 km up, laps Earth every 90 minutes, sees a tiny patch in sharp detail. Space station, Starlink. **Middle shelf (MEO):** up to 36,000 km, 12-hour laps, GPS lives here. **Top shelf (GEO):** exactly 35,786 km, one lap takes exactly one day, so from the ground it hangs still. TV and weather satellites.
Higher = sees more, moves slower, sees less detail, and the radio takes longer. Low satellites only see one spot for minutes, so to cover the planet you fly a swarm: a constellation. GPS is a swarm of 24+; your phone needs 4 overhead.

> **SAY: "Higher means you see more but move slower. One exact height hangs still, and it has to be over the equator."**

Draw: Earth, three rings, a little flashlight cone from each to show footprint size.
Friend asks "why over the equator?" → Every orbit is centered on the middle of the Earth, and the equator is the only ground circle that matches. Park it over New York and New York slides away underneath.

🌀 Scene 3 · The hula hoop

Earth is a basketball; the orbit is a hula hoop around it. To tell someone exactly how the hoop sits you need six things: **how big** (a), **how squished** (e), **how tilted** (i), **which way the tilt faces** (Ω, swivel), **which way the squished end points** (ω, spin), and **where the ball is right now** (ν). Five freeze the hoop; one moves.
Most orbits are ovals with Earth off to one side, so there's a closest point (perigee) and a farthest point (apogee). Like a swing: fastest at the bottom (perigee), slowest at the top (apogee). And those are the two spots where you fire the engine: push at perigee and the far side rises; push at apogee and the near side rises.

> **SAY: "Six numbers describe any orbit: size, squash, tilt, swivel, spin, and where you are. Fastest at the low point, slowest at the high point."**

Draw: an egg with Earth off-center, two dots labeled fast and slow.
Friend asks "why six, not three?" → Size and shape don't tell you how the hoop is turned in space. That takes three more.

🧯 Scene 4 · Five free services

Down here, the air does five jobs you never notice: it's **sunscreen** (ozone blocks harsh UV), it's **pressure** (holds gas inside materials), it's a **damper** (a buzzing ruler stops because air steals its energy), it's a **fan** (hot air rises and carries heat away), and it's a **ground wire** (static leaks away). Vacuum cancels all five. So in space: UV snaps plastics, trapped gas pushes out and fogs your lenses (outgassing), vibrations never die, hot spots stay hot (only radiators work), and charge builds until it sparks. And it all starts on the ground: clean rooms, because one speck can kill a sensor.

> **SAY: "Earth's air is doing five jobs for you right now. Space does none of them, so the spacecraft has to do all five itself."**

Draw: a box labeled spacecraft, five arrows pointing in, each labeled with what's missing.
Friend asks "so why does the ISS have those big white panels?" → Radiators. It makes its own heat and there's no air to blow it away, so it glows it off into space.

Scenes 5–7: the air bites, the fuel bill, the free lunch

🌡 Scene 5 · The air bites back

The air is a layer cake. Bottom layer (troposphere, 0–11 km): weather, planes, gets colder going up. Next (stratosphere, to 50 km): gets *warmer*, because the ozone is here soaking up UV and turning it into heat. Next (mesosphere, to 90 km): coldest, meteors burn. Then the thermosphere (to 500 km): the thermometer says 1000 °C but there's almost no air, so nothing can hand that heat to you. The space station flies through it and doesn't melt. Temperature is how fast molecules move; heat is how many there are to hit you.

Up there UV also snaps oxygen pairs into single oxygen atoms, which are desperate to grab onto anything. The spacecraft plows into them at 7.5 km/s and they eat its plastics like sandpaper. Metal shrugs; Kapton film gets shredded. Fix: coatings.

> **SAY: "The top of the atmosphere is 1000 degrees but can't burn you, because there's nothing there. What is there is loose oxygen atoms that eat plastic."**

Draw: a stack of five bands with a zigzag temperature line beside it.

Friend asks "1000 degrees and it doesn't melt?" → One sparkler is 1000 degrees and doesn't warm a room.

💸 Scene 6 · The fuel bill

First, the ladder of ways to visit a planet: fly past it, orbit it, drop a probe into its air, land on it, drive on it, or crash into it on purpose. Each rung costs more fuel to slow down and brings back more science.

Now, getting there. You don't aim at Mars; you aim at where Mars will be. The cheapest route is a lazy half-ellipse that drifts outward and meets Mars halfway around the Sun: the **Hohmann transfer**, about 8.5 months, two engine burns. You could go faster, but every bit faster costs wildly more fuel, and past a point the rocket can't be built. **Lambert's problem** is the calculator: two points plus a time, out comes the orbit and the fuel bill. And you can only launch when Earth and Mars line up, every 26 months.

> **SAY: "Fast is expensive, slow is cheap. The lazy throw takes eight and a half months and you can only do it every two years."**

Draw: two circles around a dot, a dashed half-oval connecting them.

🚚 Scene 7 · The free lunch

For really far places the lazy throw is still too expensive. So you cheat. Swing your spacecraft close past a planet. In the planet's eyes you come in and go out at the same speed, just bent. But the planet is moving around the Sun, and the bend lines you up to carry some of that motion along. You leave faster than you arrived, for free. That's a **flyby** or **gravity assist**: a ball bouncing off the front of a moving truck. The planet pays for it by slowing down an amount nobody could ever measure.

More speed means a bigger orbit, which means you can reach farther. Pass in front of the planet instead of behind and you slow down, which is how you brake at Mercury. The price is time: Rosetta took 10 years and four flybys to reach a comet. New Horizons took 9.5 years to Pluto with one Jupiter boost, and arrived too fast to ever stop, which is why it was a flyby.

> **SAY: "Bounce off a moving planet and you leave faster than you came. It's free, but it takes years."**

Draw: a planet with a thick arrow showing it moving, a dashed path bending around it.

Friend asks "isn't that free energy?" → No, it comes from the planet. Jupiter slowed down by about a trillionth of nothing.

🎯 The landing line (end on this)

"So that's the whole thing. Space gives you nothing for free. Getting there is falling and missing. Staying alive means doing the air's five jobs yourself. And going anywhere is a trade between time and fuel, unless you can borrow a planet. Every spacecraft ever built is just a different answer to those trades."

> **Then stop talking. Let them ask.**

🔁 Where this came from

Scenes 1, 2, 6, 7 = Section 1 + Section 3. Scenes 4, 5 = Section 2. Scene 3 = Section 3. Your notebook already has every fact on this page; what's new here is the order and the connecting sentences.

Minute by minute: key line, napkin drawing, hand-off

Not a script to memorize word for word. Memorize the bold key line for each block and the drawing. Everything else you say in your own words. Total: 420 seconds.

0:00 0:45 **HOOK.** "Did you know the space station is falling right now? It just keeps missing." Ball off a mountain, three throws. **Key line:** an orbit is falling and missing. *Hand-off:* "So how high you go changes everything…"

0:45 1:45 **THREE SHELVES.** Low shelf 90 min, sharp, tiny view. Middle shelf GPS 12 h. Top shelf one day, hangs still, 35,786 km, over the equator. Higher = sees more, slower, blurrier. Swarms for low orbit. **Key line:** one exact height hangs still. *Hand-off:* "But how do you even describe an orbit to someone?"

1:45 2:45 **HULA HOOP.** Six things: size, squish, tilt, swivel, spin, where the ball is. Egg with Earth off-center: fastest at the low point, slowest at the high point, fire the engine at those two spots. **Key line:** six numbers, five freeze the hoop, one moves. *Hand-off:* "Okay, you're up there. Now space tries to kill you."

2:45 3:45 **FIVE FREE SERVICES.** Sunscreen, pressure, damper, fan, ground wire. Vacuum cancels all five: UV snaps plastic, outgassing fogs lenses, vibrations never die, hot spots stay hot, static sparks. Clean rooms. **Key line:** the air does five jobs for you; space does none. *Hand-off:* "And the little bit of air that IS up there is worse."

3:45 4:30 **THE AIR BITES.** Layer cake, zigzag temperature. Top layer 1000 °C but empty: a sparkler. Loose oxygen atoms eat plastic at 7.5 km/s. **Key line:** hot but empty, and the oxygen is hungry. *Hand-off:* "Now let's leave Earth entirely."

4:30 5:45 **THE FUEL BILL.** Ladder: fly past, orbit, probe, land, drive, crash. Don't aim at Mars, aim where it'll be. Lazy half-ellipse, 8.5 months, two burns, every 26 months. Lambert = the "by Tuesday?" calculator. **Key line:** fast is expensive, slow is cheap. *Hand-off:* "Unless you cheat."

5:45 6:30 **THE FREE LUNCH.** Bounce off the front of a moving truck. Planet's view: same speed, bent. Sun's view: faster. Rosetta 10 years, New Horizons 9.5. Pass in front to brake. **Key line:** free speed, but it costs years. *Hand-off:* "So here's the whole thing in one breath."

6:30 7:00 **LANDING.** "Space gives you nothing for free. Getting there is falling and missing. Staying alive is doing the air's five jobs yourself. Going anywhere is time versus fuel, unless you borrow a planet." Then stop and let them ask.

⚡ Running long? Cut Scene 5 (the air) to one sentence: "and the thin air that's left eats plastic." Running short? Add the ISS-radiator question to Scene 4. Never cut the hook or the landing.

Do it cold. Then check page 6.

Three kinds of question, mixed: facts (F), explain-why (W), and "a friend asks" (A). The W and A ones are the ones that matter for the 7 minutes. Score yourself honestly.

1. (W) In one sentence, why doesn't an orbiting satellite fall down?
2. (F) LEO, MEO, GEO: altitude band and lap time for each.
3. (W) Why is GEO at 35,786 km and not a round number?
4. (A) "Why can't a TV satellite hang over New York?"
5. (F) Geosynchronous vs geostationary: which one hangs over one spot?
6. (W) Why does GPS need 24+ satellites when a TV satellite needs 1?
7. (F) Name the six orbital parameters and what each one answers in one word.
8. (W) Where is a satellite fastest, and which two spots do you fire the engine at? What does each burn do?
9. (F) a = ? (formula, and measured from where?)
10. (F) e = 0 means? e near 1? e over 1?
11. (A) "What does the air do for us that space doesn't?" Name all five.
12. (W) Why is a vibration a bigger problem on a spacecraft than on a car?
13. (F) What is outgassing, and why keep outgassing materials away from sensors?
14. (A) "The thermosphere is 1000 °C. Why doesn't the ISS cook?"
15. (W) Why does the stratosphere get warmer as you go up?
16. (F) What is atomic oxygen, where does it come from, and where is it the main gas?
17. (W) Who's moving at 7.5 km/s in an atomic-oxygen hit, and why does it matter?
18. (F) Rank the six "visit a planet" spacecraft types from least to most committed.
19. (W) Why can't we fly to Mars in 20 days even though an orbit for it exists?
20. (F) Describe a Hohmann transfer: shape, angle, number of burns, time to Mars.
21. (F) Lambert's problem: what goes in, what comes out?
22. (A) "Isn't a gravity assist free energy?"
23. (W) In the planet's frame, what changes during a flyby? In the Sun's frame?
24. (A) "Why did Rosetta take 10 years?" and "Why couldn't New Horizons stop at Pluto?"

Scoring: 20+ solid → you're ready to rehearse. 15–19 → rehearse, but redo the missed ones first. Under 15 → one more review pass on the weak scene before you rehearse. Every miss becomes a question-bank card.

Answers, and which scene to review if you missed it

1. It is falling; it's just moving sideways so fast the ground curves away as fast as it falls. *(Scene 1)*
2. LEO 200–2,000 km, 1.5–2 h. MEO 2,000–35,786 km, 2–24 h (GPS 12 h). GEO 35,786 km, 24 h. *(Scene 2)*
3. It's the one altitude where one lap takes exactly one Earth rotation. Calculated, not chosen. *(Scene 2)*
4. Every orbit is centered on Earth's center; only the equator moves in a circle an orbit can match. Over New York it would drift north and south daily. *(Scene 2)*
5. Geostationary (one-day lap AND over the equator). Geosynchronous = one-day lap at any tilt. *(Scene 2)*
6. GPS is low (MEO), so each satellite sees a limited patch and moves; you need many so 4 are always overhead. A GEO satellite sees a third of Earth and never moves. *(Scene 2)*
7. a size, e shape, i tilt, Ω swivel, ω spin, ν position. *(Scene 3)*
8. Fastest at perigee. Burn at perigee → apogee rises; burn at apogee → perigee rises. *(Scene 3)*
9. a = (r_perigee + r_apogee) ÷ 2, both measured from Earth's center. *(Scene 3)*
10. 0 = circle. Near 1 = long cigar. Over 1 = open path that never closes (a flyby). *(Scene 3)*
11. Sunscreen (ozone vs UV), pressure, damper, fan (convection), ground wire (discharge). *(Scene 4)*
12. On Earth the air absorbs the vibration and it dies out. In vacuum nothing does; the spacecraft must damp it itself. *(Scene 4)*
13. Trapped gas leaking out of materials in vacuum. It condenses on cold surfaces, usually lenses and sensors, and fogs them. *(Scene 4)*
14. Almost no molecules up there. Temperature is molecule speed; heat needs numbers. A sparkler is 1000 °C and won't warm a room. *(Scene 5)*
15. The ozone layer lives there and absorbs UV; absorbed light becomes heat. *(Scene 5)*
16. Single O atoms made when UV splits O_2. Main gas from ~200–600 km, exactly where LEO satellites fly. *(Scene 5)*
17. The spacecraft. The atoms are nearly still; the spacecraft slams into them, so each hit carries enough energy to break a chemical bond. *(Scene 5)*
18. Flyby → orbiter → atmospheric probe → lander → rover → penetrator. Each step costs more fuel/risk, returns more science. *(Scene 6)*
19. The Δv needed is absurd; no rocket could carry the fuel. Fast is expensive. *(Scene 6)*
20. Half an ellipse touching both orbits; depart and arrive 180° apart; two burns; ~8.5 months to Mars. *(Scene 6)*
21. In: two positions + a travel time. Out: the one orbit connecting them, and therefore the Δv. *(Scene 6)*
22. No. The energy comes from the planet, which slows down by an unmeasurably tiny amount. *(Scene 7)*
23. Planet's frame: speed unchanged, direction bends. Sun's frame: speed changes because you add the planet's own motion. *(Scene 7)*
24. Rosetta used four flybys to build speed for free instead of carrying fuel; time was the price. New Horizons left Earth so fast (plus a Jupiter boost) that no fuel could brake it into Pluto orbit. *(Scene 7)*

Three runs, then do it for real

🎬 Three rehearsal runs

- **Run 1, with page 4 open.** Talk through all seven blocks, reading the key lines. Don't time it. Goal: feel the order.
- **Run 2, with only the cue card** (right). Phone timer running. Expect to run long. Note which block ate the time.
- **Run 3, nothing in front of you, phone recording.** Timer running. Then listen back and mark: every "um," every sentence that ended in a word you didn't explain, every place you reached for a fact and it wasn't there. Those three things are your fix list.
- **Then send me the recording or a transcript.** I'll grade it the way the friend would hear it, not the way a textbook would.

🧯 When you get stuck mid-sentence

- Pivot to the picture: "The way I think about it is…" then the analogy for that block (ball off a mountain, three shelves, hula hoop, five jobs, sparkler, lazy throw, moving truck).
- Forgot a number? Say the idea instead. "About a day" beats a frozen pause over "23 hours 56 minutes."
- Lost your place? Go to the hand-off line of the block you're in. They all start with "So…" or "But…"

📏 What "holding your own" sounds like

- You use the analogy first and the real word second: "it's like a hula hoop around a basketball, the six orbital parameters…"
- You can answer one "why" past each fact. Not two. One is enough for a friend.
- You say "I'm not sure, but I think…" for the things you're not sure about. That's what people who actually know things sound like.

🃏 Cue card (copy onto one index card)

1 · FALL & MISS — ball off a mountain
2 · 3 SHELVES — 90 min / 12 h / 1 day · 35,786 · equator · swarm
3 · HULA HOOP — size squash tilt swivel spin bead · fast low, slow high · burn there
4 · 5 FREE JOBS — sunscreen pressure damper fan ground wire · clean room
5 · AIR BITES — layer cake zigzag · 1000° sparkler · hungry oxygen 7.5 km/s
6 · FUEL BILL — ladder · lazy throw 8.5 mo · every 26 mo · fast = $$$
7 · FREE LUNCH — moving truck · Rosetta 10 yr · NH 9.5 yr
LAND — "space gives you nothing for free"

Eight lines. If you can talk for 7 minutes from these eight lines, you own it.

⚡ Order of work today: read pages 2–3 once → do the quiz cold (page 5) → check (page 6) → fix the misses → Run 1 → Run 2 → Run 3 → send me the recording.